Made by: Jason Kenneth Fo & Matthew Jovan Widjaja | CS-A (Pak Daniel)

Sumber: https://www.idx.co.id/id/perusahaan-tercatat/laporan-keuangan-dan-tahunan/

Goal: To analyze and forecast Sales using historical Sales data for a manufacturing company, applying the time series analysis method.

Study Case: As a Sales department in charge of one of the manufacturing companies, your team will identify its historical Sales data from 2021 to Q2 2025 (18 data points).

Time Series Analysis of *Indocement Tunggal Prakarsa Tbk* (INTP) | Q1 - Q4 of 2021 - 2025

Year/Period	Saless (in IDR, Rp 1 000 000)	Three-quarter moving average (in IDR, Rp 1 000 000)	Four-quarter moving average (in IDR, Rp 1 000 000)	Projected Sales 2025/Q3 Three-quarter moving average (in IDR, Rp 1 000 000)	Projected Sales 2025/Q4 Three-quarter moving average (in IDR, Rp 1 000 000)	Projected Sales 2025/Q3 Four-quarter moving average (in IDR, Rp 1 000 000)	Projected Sales 2025/Q4 Four-quarter moving average (in IDR, Rp 1 000 000)	Variations based on Three-quarter moving average (in IDR, Rp 1 000 000)	Variations based on Four-quarter moving average (in IDR, Rp 1 000 000)	Average Cyclical Variations based on Three-quarter moving average (in IDR, Rp 1 000 000)	Average Cyclical Variations based on Four-quarter moving average (in IDR, Rp 1 000 000)	Average Seasonal Variations based on Three-quarter moving average for Q3 (in IDR, Rp 1 000 000)	Average Seasonal Variations based on Four-quarter moving average for Q3 (in IDR, Rp 1 000 000)	Average Seasonal Variations based on Three-quarter moving average for Q4 (in IDR, Rp 1 000 000)	Average Seasonal Variations based on Four-quarter moving average for Q4 (in IDR, Rp 1 000 000)	Adjusted Projected Sales 2025/Q3 Three-quarter moving average (in IDR, Rp 1 000 000)	Adjusted Projected Sales 2025/Q4 Three-quarter moving average (in IDR, Rp 1 000 000)	Adjusted Projected Sales 2025/Q3 Four-quarter moving average (in IDR, Rp 1 000 000)	Adjusted Projected Sales 2025/Q4 Four-quarter moving average (in IDR, Rp 1 000 000)
2021/Q1	3,438,045.00																			
2021/Q2	3,228,895.00	3,536,217.67						-307,322.67												
2021/Q3	3,941,713.00	3,777,953.67	3,707,936.75					163,759.33	233,776.25											
2021/Q4	4,163,253.00	3,887,564.33	3,738,453.75					275,688.67	424,799.25											
2022/Q1	3,557,727.00	3,691,443.00	3,855,030.00					-133,716.00	-297,303.00											
2022/Q2	3,353,349.00	3,886,981.67	4,019,059.50					-533,632.67	-665,710.50											
2022/Q3	4,749,869.00	4,256,850.33	4,168,063.25					493,018.67	581,805.75											
2022/Q4	4,667,333.00	4,554,293.00	4,300,430.88					113,040.00	366,902.13											
2023/Q1	4,245,677.00	4,212,450.00	4,372,780.38					33,227.00	-127,103.38											
2023/Q2	3,724,340.00	4,309,230.33	4,443,097.50					-584,890.33	-718,757.50											
2023/Q3	4,957,674.00	4,568,026.33	4,467,050.38					389,647.67	490,623.63											
2023/Q4	5,022,065.00	4,687,435.67	4,486,390.50					334,629.33	535,674.50											
2024/Q1	4,082,568.00	4,382,267.67	4,555,902.00					-299,699.67	-473,334.00											
2024/Q2	4,042,170.00	4,440,224.67	4,611,434.13					-398,054.67	-569,264.13											
2024/Q3	5,195,936.00	4,822,055.33	4,623,828.50					373,880.67	572,107.50											
2024/Q4	5,228,060.00	4,799,908.00	4,612,338.25					428,152.00	615,721.75											
2025/Q1	3,975,728.00	4,420,292.00						-444,564.00												
2025/Q2	4,057,088.00																			





